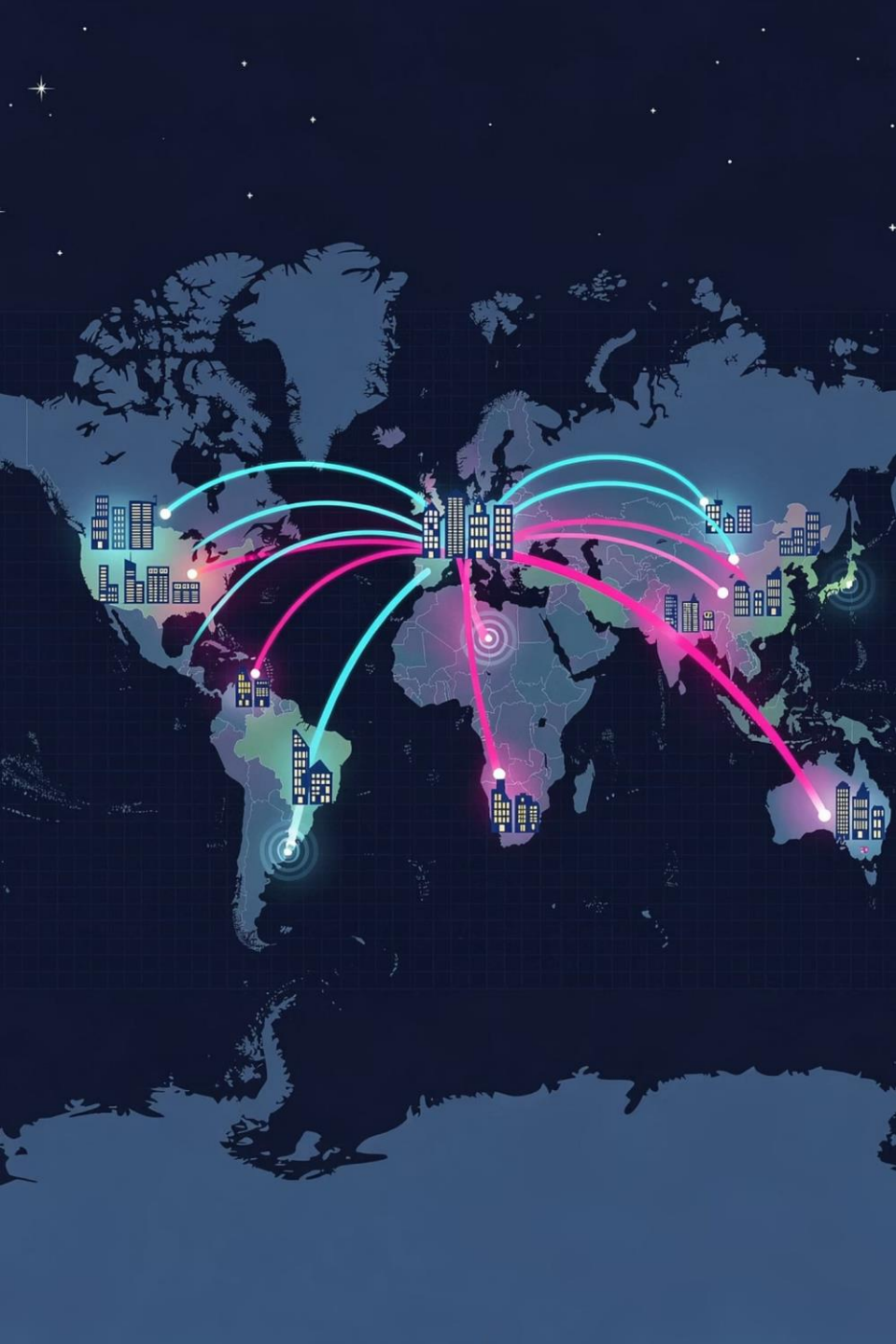




DNS & Global Delivery Networks

Route 53 & CloudFront CDN · ITNW 1009

Today we answer two questions every Cloud Support Associate faces on the job: **How does a customer's computer find our servers anywhere on Earth?** And once they find us, **how do we deliver content instantly without making them wait?** By the end of this session, you'll understand Amazon Route 53 and Amazon CloudFront – two tools that power the global internet experience.

What We're Covering



DNS Resolution Mechanics

How a domain name becomes an IP address — step by step through Route 53.



Content Delivery Networks

How Amazon CloudFront eliminates latency with edge caching worldwide.



Route 53 Routing Policies

Weighted, Latency-Based, Failover, and Geolocation — smart traffic management.



Cache Control & Security

TTL, invalidations, Origin Access Control, and SSL/TLS certificates.



Translating Names to Numbers

Computers communicate using binary IP addresses like `192.0.2.45`. Humans rely on readable domain names like `mycompany.com`. **DNS – the Domain Name System – is the Internet's global directory service**, bridging these two worlds every time you open a browser tab.

Human-Readable

`www.example.com` – easy to remember, impossible for a router to use directly.

DNS Magic

DNS resolves the name to a numeric IP address in milliseconds, invisibly.

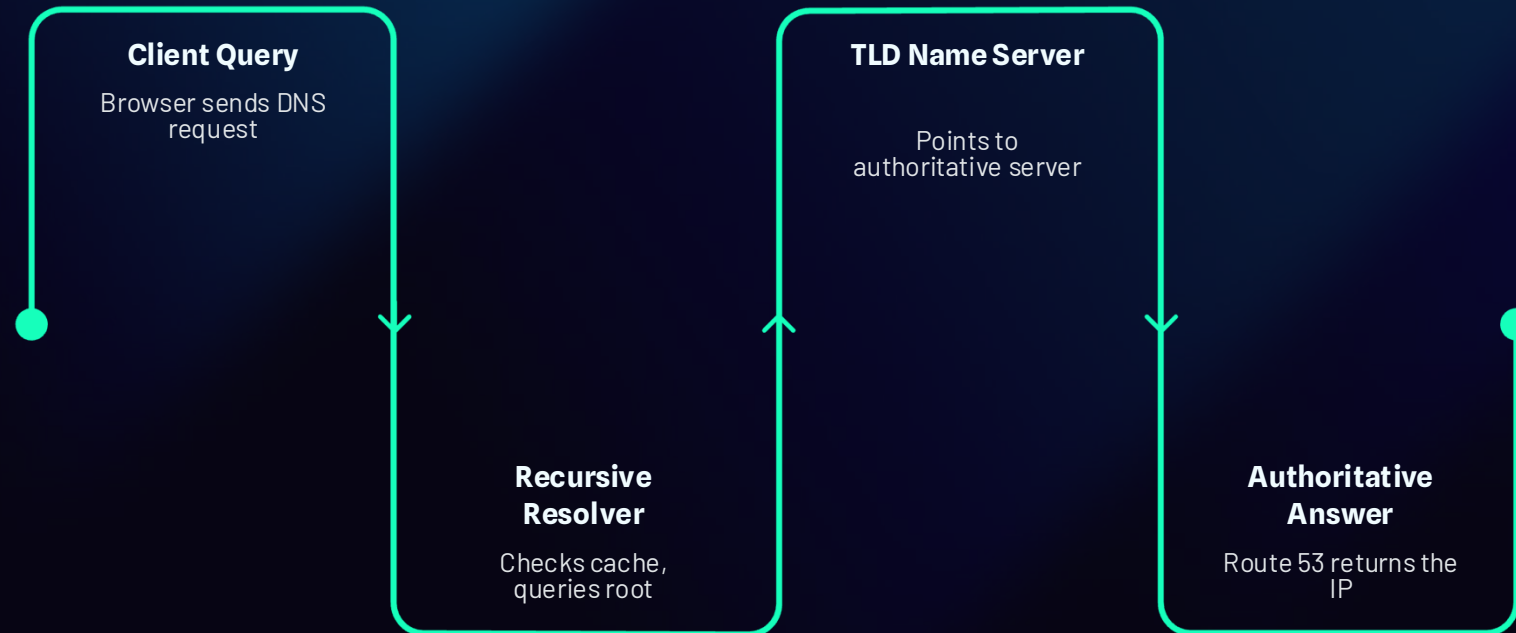
Machine-Readable

`192.0.2.45` – the actual address routers use to deliver your data packets.

❏ Think of DNS like your phone's contact list – you tap a name and it dials the number behind the scenes. Amazon Route 53 does this for the entire Internet.

The Step-by-Step DNS Resolution Journey

Every time you press Enter after typing a URL, a four-step lookup happens in milliseconds. Here's exactly how your browser finds the server it needs:



Key Players

- **Recursive Resolver:** Your ISP's server that does the searching on your behalf.
- **Root & TLD Servers:** Direct traffic based on domain endings (.com, .org, .gov).
- **Authoritative Name Server:** Holds the final, exact record — that's Route 53!

Why "Route 53"?

Port **53** is the official standard networking port where DNS operates across all networking equipment worldwide. AWS named this service Route 53 as a direct nod to that fundamental standard.

When Route 53 finds the answer in its **Hosted Zone**, it returns the IP address back through the resolver chain to your browser — connection established!

Route 53 Core Components

Hosted Zones & DNS Records

A **Hosted Zone** is your master digital ledger for a domain you own — it lives inside Route 53 and contains all the DNS records that tell the Internet where to find your resources.



A Record (Address)

Maps a domain name directly to an **IPv4 address**. Use this when pointing to a specific EC2 server: `server.mycompany.com` → `192.0.2.10`.



CNAME Record

Maps one domain name to **another domain name**. Cannot be used on root apex domains (e.g., cannot apply to `mycompany.com` directly — only subdomains).



Alias Record (AWS-Specific)

AWS's solution to the root-domain CNAME limitation. Routes traffic directly to **AWS resources** — ALBs, S3 buckets, CloudFront — even at the root apex level with zero performance penalty.



AAAA Record

Maps a domain name directly to an **IPv6 address**. The modern equivalent of the A Record for next-generation IP addressing.



Route 53 Is a Smart Traffic Cop

Route 53 doesn't just answer address questions — it routes traffic **intelligently** using configurable policies. These policies allow you to optimize for speed, safety, geography, and controlled deployments — all from a single DNS service.

Routing Policies — Part 1

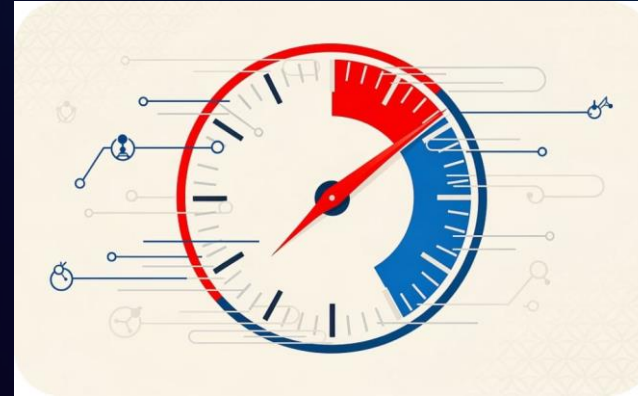
Weighted & Latency-Based Routing



Weighted Routing

Distributes traffic across multiple resources by assigned **percentage weights**. Perfect for canary testing — send 90% of traffic to your stable production server and 10% to your new version. If all looks good, dial it up gradually with zero risk.

- Use case: Blue/green deployments
- Use case: A/B feature testing
- Control: Fine-grained percentage splits



Latency-Based Routing

Measures real-time network speeds and routes **each user to the AWS Region** that offers the lowest latency for that specific user — not hardcoded geography, but dynamic live measurement. A Tokyo user goes to Asia Pacific; a Dallas user goes to US East.

- Use case: Global apps needing fast response
- Use case: Multi-region deployments
- Advantage: Network paths fluctuate — this adapts dynamically

Routing Policies — Part 2

Failover & Geolocation Routing

Failover Routing

Your ultimate safety net. Route 53 continuously sends **health checks** to your primary server. If it crashes, Route 53 detects the outage within seconds and automatically redirects all traffic to a backup server or a static S3 maintenance page — your users never see a broken screen.

- Paired with: Route 53 Health Checks
- Backup target: Secondary EC2 or S3 static page
- Recovery: Automatic, no manual intervention

Geolocation Routing

Directs traffic based on the **physical location of the user** — country, state, or continent. If a user connects from France, Route 53 sends them to your European server, serving French content and enforcing GDPR compliance automatically.

- Use case: Localized language content
- Use case: Regional licensing compliance (GDPR)
- Use case: Content restrictions by territory

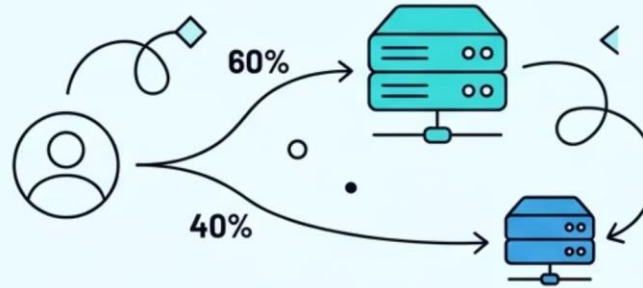
Geoproximity Routing goes further — routes based on resource geography with visual bias adjustments for fine-tuned traffic shaping.

Routing Policy Quick Reference

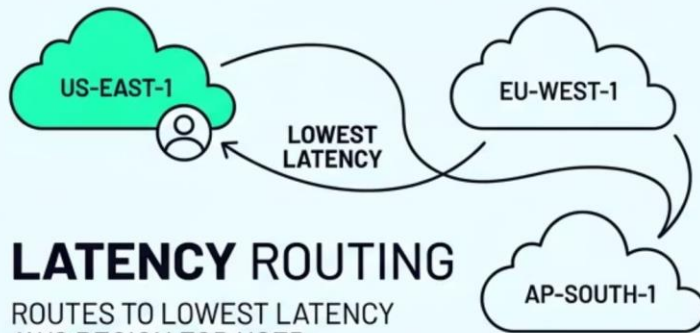


SIMPLE ROUTING

SINGLE RECORD. ONE ENDPOINT. NO LOGIC.

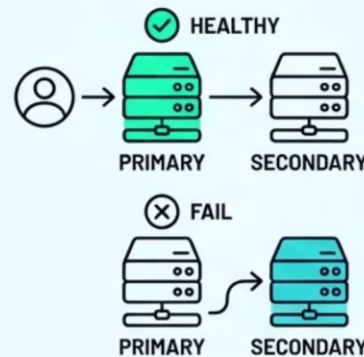


ASSIGNS PERCENTAGE WEIGHTS. TRAFFIC SPLITTING.



LATENCY ROUTING

ROUTES TO LOWEST LATENCY AWS REGION FOR USER.



FAILOVER ROUTING

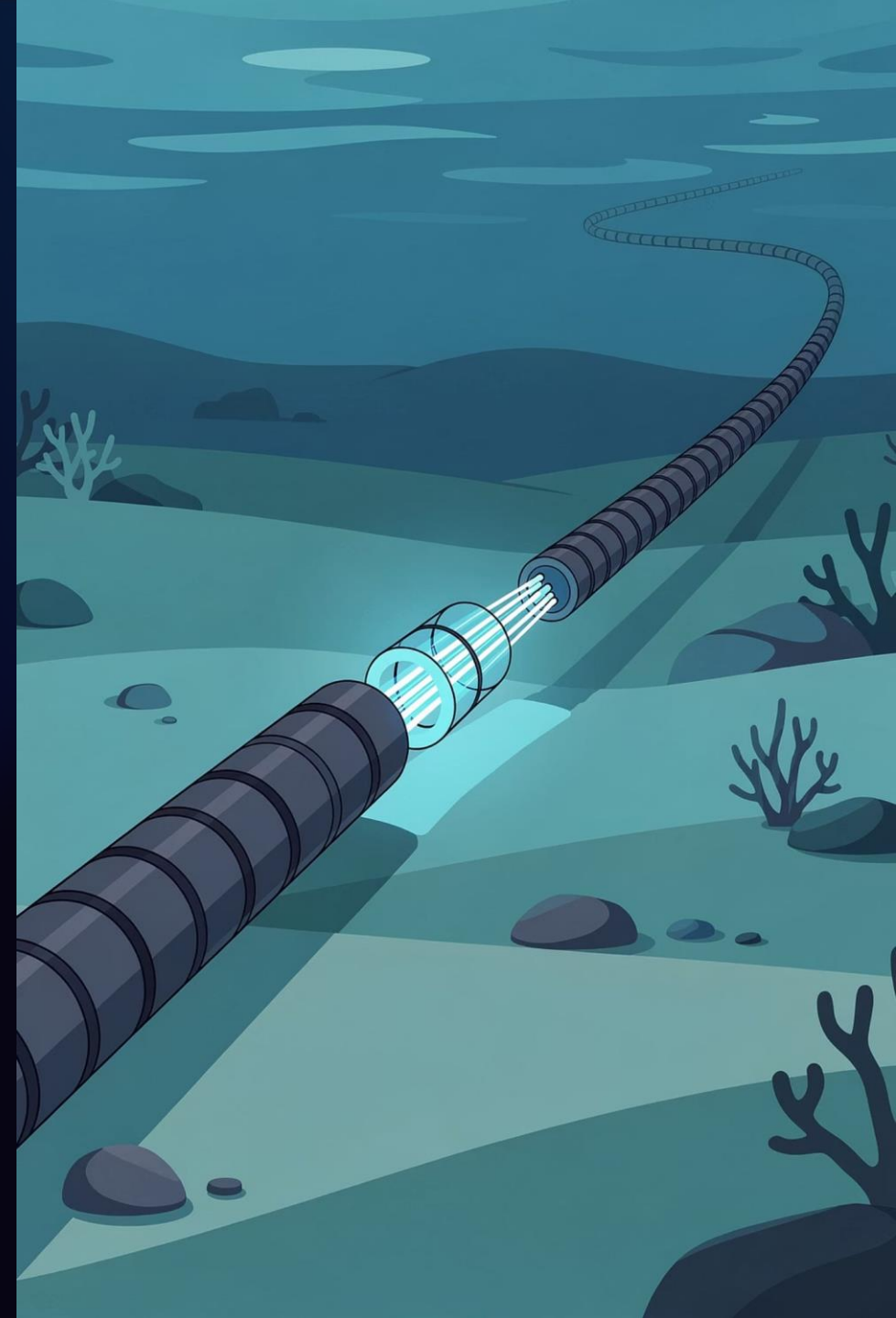
PRIMARY WITH HEALTH CHECK.

CHECK. AUTO-SWITCHES TO SECONDARY.

Choosing the right routing policy is a core skill for Cloud Support Associates. Match the policy to the business need: use **Weighted** for testing, **Latency** for global performance, **Failover** for resilience, and **Geolocation** for compliance and localization.

Physical Distance = Latency

When your server lives in Virginia and your customer lives in Sydney, data must travel thousands of miles of undersea fiber-optic cables. That physical distance causes buffering wheels, slow page loads, and frustrated customers. **Content Delivery Networks solve this problem.**



The Need for Speed — Content Delivery Networks

The Problem

A single origin server in N. Virginia serving users in Sydney, Tokyo, London, and São Paulo must respond to every request across thousands of miles of network. Even at the speed of light, round-trip times reach 200–300ms — creating a poor user experience at scale.

The Solution: CDN

A **Content Delivery Network (CDN)** is a globally distributed network of proxy servers that cache content close to end users. Instead of fetching from Virginia each time, users pull from a server in their own city.

Amazon CloudFront

AWS's low-latency, high-speed CDN service — integrated with the full AWS global infrastructure.

- **600+ Edge Locations** worldwide
- Regional Edge Caches as a second-tier buffer
- Sub-millisecond local cache delivery
- Native integration with S3, EC2, ALB, and Route 53
- Built-in DDoS protection via AWS Shield

How CloudFront Works

Origins, Distributions & Edge Caches



Origin

Your content's home base — an S3 bucket, EC2 instance, Elastic Load Balancer, or external web server.

Distribution

The CloudFront configuration set. Assigns you a global URL like `d1234.cloudfront.net`.

Cache Hit

The file is already at the local Edge Location — returned to the user instantly in milliseconds.

Cache Miss

File not at Edge. CloudFront fetches from Origin, caches locally for future requests, then delivers.

Managing the Cache

TTLs, Invalidations & Origin Security

Time-To-Live (TTL)

Controls how long files stay at the Edge before CloudFront re-checks the Origin for updates. **Default: 24 hours.** Short TTLs mean fresher content; longer TTLs mean better cache efficiency and lower origin load. Configure per object or per behavior in the distribution.

Cache Invalidation

Need to push a fix immediately – before TTL expires? Run a **CloudFront Invalidation** to force all Edge Locations worldwide to purge the stale cached file right now. Useful for urgent bug fixes, breaking news corrections, or emergency content recalls. Note: mass invalidations incur minor additional costs.

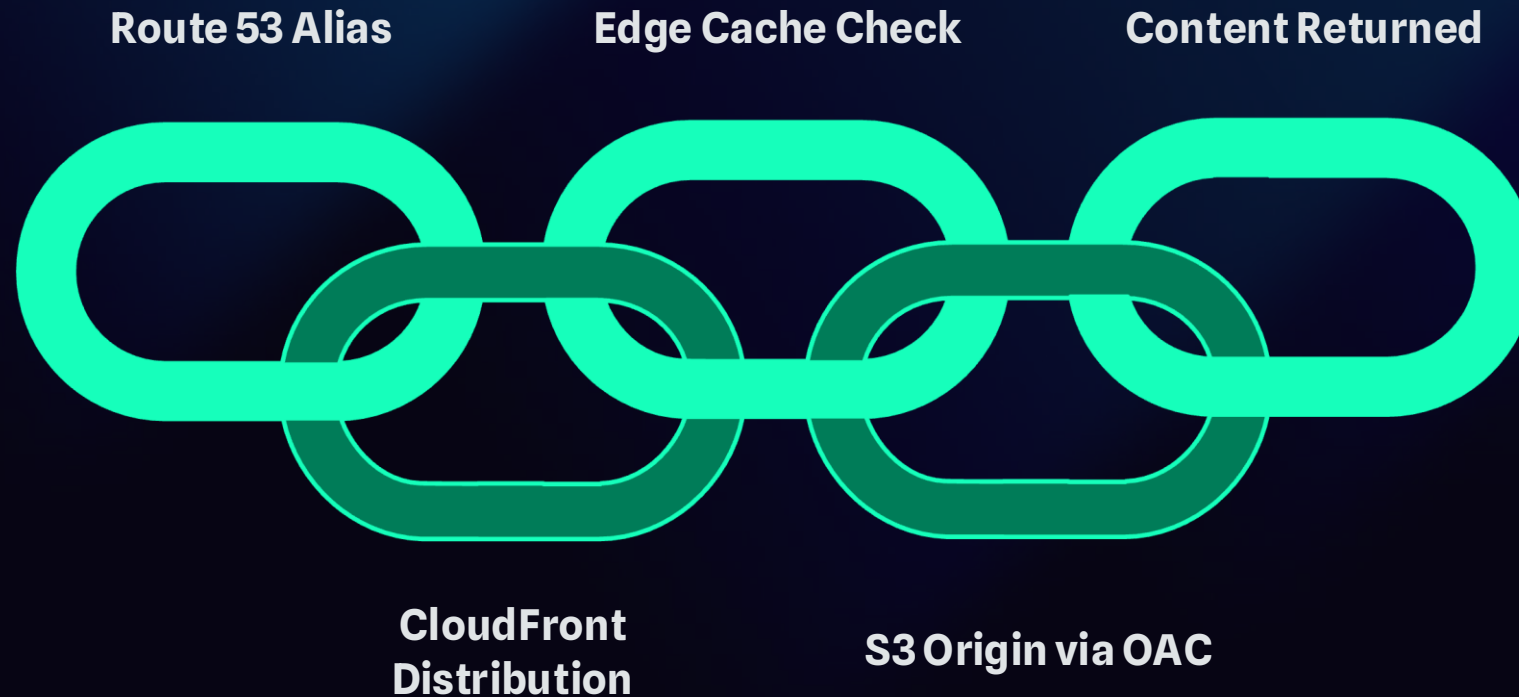
Origin Access Control (OAC)

Prevents users from bypassing CloudFront and downloading files directly from your S3 bucket. OAC locks the S3 bucket policy so that **only CloudFront** has read permission. Users who try to hit the raw S3 URL receive an Access Denied error – all traffic must flow through the CDN.

HTTPS & SSL/TLS Certificates

Enforces encrypted, end-to-end data transmission using **custom SSL/TLS certificates** provisioned through AWS Certificate Manager (ACM). Attach your own domain's certificate to your CloudFront distribution to serve traffic over `https://yourcompany.com` with full encryption.

Route 53 + CloudFront: Better Together



In production, Route 53 and CloudFront work as a team. Route 53 resolves your domain name and routes users to the CloudFront distribution endpoint using an **Alias Record** at the root apex. CloudFront then intercepts every request, serves cached content from the nearest Edge Location, and only calls back to the S3 or EC2 origin on cache misses — dramatically reducing latency and origin load.

Real-World Analogy

The International Newsstand



Route 53 = Directory Assistance

You call and say the publication name (domain). The operator looks in the master ledger (Hosted Zone), gives you routing directions (IP address), and applies smart rules — if the main office is closed, they give you the backup address (Failover); if you speak French, the Paris office (Geolocation).



S3 Origin = Central Warehouse

The publishing warehouse in Atlanta holds the master copies. If every reader worldwide had to fly to Atlanta to pick up a newspaper, travel would grind to a halt. That's your single-region origin server under heavy global load.



Edge Locations = Local Newsstands

Thousands of neighborhood newsstands (Edge Locations) hold local copies. The first London reader triggers a shipment from Atlanta (Cache Miss). Every reader after that grabs a copy off the shelf instantly (Cache Hit) — no transatlantic delay.



TTL = Newspaper Expiration Date

Each paper is stamped "Discard after 24 hours" (TTL). If a major story breaks and the morning print is wrong, the publisher sends an urgent recall to all shelves immediately (Cache Invalidation) — clearing the error before more readers see it.

Analogy 2

The Global Restaurant Chain

1

Central Hotline

One global phone number — Route 53 dispatches your order to the best kitchen using smart routing policies.

2

Local Delivery Hub

Neighborhood warming hubs (Edge Locations) pre-stage popular menu items. No need to wait for the central vault every time.

3

Secure Vault

The central ingredient warehouse (S3 Origin) is locked — only authorized delivery drivers (CloudFront via OAC) can collect from it.

4

Tamper-Evident Bag

Food travels in sealed, insulated bags (SSL/TLS encryption) — no one can tamper with or inspect your order in transit.

Every element of CloudFront's security and delivery model maps directly to something you already understand intuitively — use these mental models when working through exam scenarios or on-the-job architecture reviews.

Key Vocabulary



DNS

Domain Name System – the global directory that translates human-readable domain names into machine-readable IP addresses.



Route 53

Amazon's highly available, scalable managed DNS and traffic routing service. Named after Port 53.



CloudFront / CDN

Amazon's Content Delivery Network – caches content at 600+ global Edge Locations to minimize latency.



Edge Location

A CloudFront data center positioned close to end users to serve cached content with minimal delay.



TTL

Time-To-Live – how long a cached file stays at an Edge Location before CloudFront checks for updates.



OAC

Origin Access Control – restricts S3 bucket access so only CloudFront can read files, blocking direct public access.

Active Learning — 25 Minutes

Global Latency & Routing Policy Workshop

You are a **Junior Cloud Support Engineer at GlobeFlix**, an international video streaming platform. Your S3-hosted video library lives in US East (N. Virginia). Application servers run in US East and EU (Frankfurt). Three customer complaints just landed on your desk — your job is to prescribe the correct AWS solution for each.

- i** Format: Think-Pair-Share (10 min pair analysis → 10 min group debate → 5 min instructor debrief). Write your reasoning on your worksheet before sharing with the group.



GlobeFlix — The Three Issues

1

Issue A — Sydney Slowdown

Users in Sydney complain that browsing static thumbnails and downloading promotional video trailers takes **8–12 seconds per page load**. What AWS service solves this?

2

Issue B — Checkout Split Test

The engineering team is launching a redesigned checkout page. They want **15% of real traffic** to test the new page while 85% stays on the proven old page. Which Route 53 policy?

3

Issue C — EU Crash Page

The primary European database server occasionally crashes during peak football matches. European users see a **raw 502 Bad Gateway error** instead of a friendly maintenance page. How do you fix this?

Model Answers — GlobeFlix Workshop

Issue A

Deploy an **Amazon CloudFront Distribution** in front of the S3 bucket. Static thumbnails and trailers cache at Australian Edge Locations, reducing load from 8–12 seconds down to milliseconds. No new servers needed in Australia.

Issue B

Configure a **Route 53 Weighted Routing Policy**. Assign weight 85 to the original checkout page record and weight 15 to the new checkout page record. Route 53 distributes DNS resolutions accordingly across the global user base.

Issue C

Implement a **Route 53 Failover Routing Policy** paired with Route 53 Health Checks. Primary record points to the EU server. Secondary (failover) record points to a static maintenance page hosted on S3 — activated automatically on server failure.

📌 Key insight on Issue B: Weighted routing applies across all new DNS resolutions globally — not per individual browser. A user who refreshes 10 times won't toggle predictably between versions because DNS responses are cached locally by TTL.

Vocational Case Study

Streaming Under Fire — The GlobeNews Crisis

In November 2024, **GlobeNews 24** covered a sudden high-stakes international breaking news event. In twenty minutes, concurrent viewers spiked from 25,000 to **over 1.8 million global streaming connections** — with zero CDN protection in place. What happened next is a textbook lesson in why CloudFront matters.



The Cascade Failure

What Went Wrong Without CloudFront

The Architecture Flaw

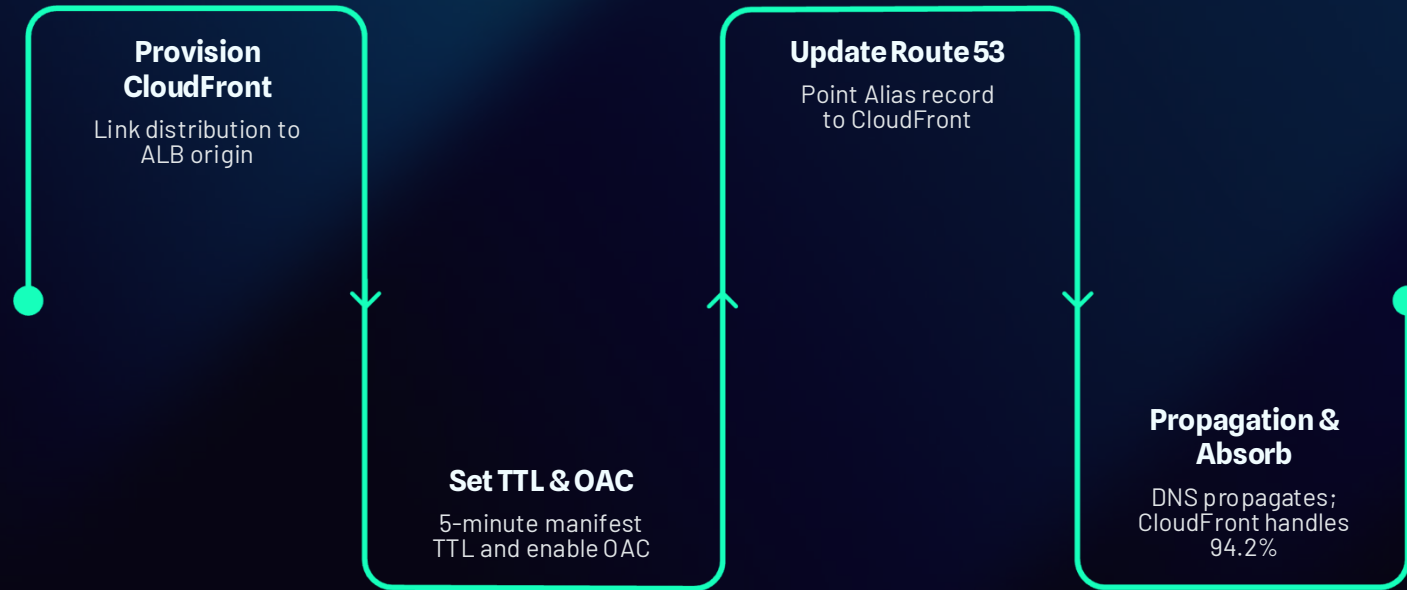
GlobeNews routed all user traffic via Route 53 Simple Routing directly to an Application Load Balancer and EC2 fleet in N. Virginia. **Every browser worldwide repeatedly requested homepage assets, video manifests, and breaking news graphics directly from origin servers** — no caching, no edge distribution.

The Impact

- EC2 CPU utilization hit **100%** within 30 minutes
- Auto Scaling could not launch instances fast enough
- HTTP 504 Gateway Timeouts for international viewers
- Infinite buffering loops across Europe, Asia, South America
- AWS Data Transfer Out costs **skyrocketed exponentially**

The Emergency Remediation

Six Minutes to Recovery



94.2%

Cache Hit Ratio

CloudFront absorbed nearly all incoming requests at Edge Locations – origin servers barely touched.

60%

Bandwidth Cost Drop

CloudFront's discounted edge transfer rates slashed outbound bandwidth costs by over 60 % immediately.

18%

EC2 CPU After Fix

Down from 100% CPU exhaustion to a healthy 18% utilization within minutes of DNS propagation.

150ms

Global Stream Latency

Stabilized under 150ms worldwide after CloudFront Edge Locations distributed the load globally.

Case Study Debate Questions

1 Architectural Analysis

Why did routing traffic through an Elastic Load Balancer directly to an Auto Scaling group fail during a massive global traffic spike — and how did introducing CloudFront Edge Locations resolve the CPU exhaustion at the origin?

2 Financial & Cost Optimization

How does caching static and media assets at a CloudFront Edge Location impact a company's monthly AWS bill compared to serving those same assets directly out of EC2 or Amazon S3? What are the specific pricing differences?

3 Operational Security & Control

If an emergency breaking news update requires a corrected headline graphic, what operational problem arises from CloudFront's edge cache — and what **two technical solutions** can engineers use to fix it immediately?

Case Study Grading Checklist

- REQUIREMENT 1: Student explains that ALB/EC2 Auto Scaling failed because every request required full server-side processing, network stack allocation, and long-distance transmission across continents.
- REQUIREMENT 1B: Student identifies that CloudFront's 94.2% Cache Hit Ratio converted most origin calls into fast local edge responses, preventing CPU overload entirely.
- REQUIREMENT 2: Student correctly states that EC2/ALB Data Transfer Out is charged at higher standard Internet rates vs. CloudFront's tiered, discounted edge transfer pricing.
- REQUIREMENT 2B: Student notes that CloudFront-to-Origin data transfer within AWS is free, significantly reducing Total Cost of Ownership (TCO).
- REQUIREMENT 3A: Student proposes Solution A – execute a CloudFront Invalidation request to immediately purge the specific file path (e.g., /images/headline.png) across all global Edge Locations.
- REQUIREMENT 3B: Student proposes Solution B – implement Object Versioning / Cache Busting by renaming the updated file (e.g., headline_v2.png) and updating the HTML reference, avoiding invalidation costs entirely.

Key Takeaways

DNS Translates Names

Route 53 resolves human-readable domain names to IP addresses through a 4-step resolver chain, culminating at the Authoritative Name Server in your Hosted Zone.

Routing Policies Are Your Toolkit

Weighted = testing. Latency = global performance. Failover = resilience. Geolocation = compliance and localization. Match policy to business need.

CloudFront Eliminates Distance

By caching content at 600+ Edge Locations, CloudFront turns cross-continental round-trips into millisecond local cache hits – reducing latency, origin load, and bandwidth cost simultaneously.

Security Layers Are Non-Negotiable

OAC locks S3 so only CloudFront can read it. SSL/TLS certificates encrypt transit. TTLs balance freshness vs. efficiency. Invalidations give you emergency override control.